

Macroeconomic Shocks and Equity Market Dynamics in Vietnam: A VAR Analysis

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Abstract – This paper examines the impact of macroeconomic shocks on VN-Index returns from January 2010 to December 2024 using a six-variable VAR(2) model with structural identification based on Cholesky decomposition. The results show that U.S. monetary policy, proxied by the Federal Funds Rate, is the dominant external driver of Vietnamese equity market dynamics, while domestic macroeconomic variables play a comparatively limited role. Variance decomposition indicates that VN-Index fluctuations are largely driven by idiosyncratic shocks, highlighting the importance of market-specific factors. Overall, the findings suggest that VN-Index dynamics are primarily shaped by global financial conditions rather than domestic macroeconomic fundamentals.

Keywords: VN-Index, Vector Autoregression, Impulse Response Function, Variance Decomposition, Federal Funds Rate, Emerging Markets.

I. INTRODUCTION

Stock markets serve as a critical conduit for capital formation and resource allocation, linking savers with productive investment opportunities while also reflecting broader macroeconomic conditions. In emerging economies, where financial systems are still developing, equity markets play an especially important role in mobilising long-term capital for both firms and governments. In Vietnam, the VN-Index – the benchmark index of the Ho Chi Minh Stock Exchange (HOSE) – has expanded substantially since its establishment in 2000, accompanied by rapid growth in listed firms, market capitalisation, and investor participation. According to statistics released by the Ministry of Finance and the State Securities Commission of Vietnam, Vietnam’s stock market capitalisation reached approximately USD 247 billion by the end of 2023, equivalent to roughly 62% of

GDP, while the number of securities accounts approached 7.4 million, representing around 7.5% of the population. Beyond its financing role, the VN-Index also functions as a real-time indicator of macroeconomic sentiment. The U.S. Federal Reserve tightening cycle of 2022–2023 illustrated the market’s pronounced sensitivity to external monetary conditions, with VN-Index declining sharply as global financial conditions tightened. Understanding the macroeconomic drivers of VN-Index fluctuations is therefore important for investors managing systematic risk, policymakers monitoring financial stability, and researchers studying shock transmission in an emerging open economy.

The macroeconomic approach to equity pricing is based on the Arbitrage Pricing Theory (APT) of [1] and empirically developed by [2], who show that systematic macroeconomic variables, rather than a single market factor, drive equity returns. Within this framework, asset prices reflect expect-

tations about future economic activity, implying that any systematic, non-diversifiable risk factor should be priced into equity returns. [2] identify five main transmission channels. First, interest rate changes affect the discount rate applied to future cash flows; an increase in rates reduces the present value of expected earnings and depresses equity prices [3]. Second, inflation reduces the real value of nominal returns and increases uncertainty, with unexpected inflation playing the key role in asset pricing since expected inflation is already incorporated into nominal rates [4]. Third, exchange rate movements influence firms' international competitiveness, where currency depreciation may improve export performance in the short run but can also signal macroeconomic instability [5]. Fourth, oil price shocks generate supply-side cost pressures that reduce corporate margins and equity valuations [6]. Finally, for small open emerging economies such as Vietnam, external monetary conditions constitute an important additional channel. International factors are incorporated into equity pricing through trade and capital flow linkages even in partially segmented markets [7]. In particular, unexpected changes in the U.S. Federal Funds Rate affect global risk premium and capital flows [3], with strong spillover effects documented during recent tightening cycles [8].

Existing studies on Vietnam primarily examine the transmission of monetary policy to macroeconomic aggregates such as real GDP(output) and inflation, with relatively limited attention devoted to equity market dynamics [9]. While several studies have applied VAR-based approaches to analyse financial market interactions in Vietnam [5], the existing literature generally focuses on isolated dimensions of market behaviour rather than providing a comprehensive assessment that jointly incorporates causality analysis, dynamic shock transmission, and variance decomposition within a unified empirical framework. Despite a growing body of literature on macro-financial linkages in emerging markets, two important gaps remain.

First, existing studies on Vietnam focus predominantly on the transmission of monetary policy to output and inflation, leaving equity market dynamics comparatively underexplored. Second, to date, there appears to be no study that has jointly examined domestic and external macroeconomic shocks within a unified empirical framework that simultaneously addresses causality, dynamic shock propagation, and variance decomposition for the Vietnamese equity market. This paper addresses both gaps by employing a VAR framework over the period 2010–2024, a sample that encompasses significant shifts in both domestic and global financial conditions.

This paper contributes to the VAR-based literature on macro-financial dynamics in emerging equity markets in three ways. First, it provides new evidence on the relative importance of domestic versus external macroeconomic shocks in driving VN-Index dynamics over 2010–2024, explicitly incorporating U.S. monetary policy conditions – proxied by the Federal Funds Rate – as a key external driver alongside domestic variables. This extends the existing Vietnam-focused literature, which has largely overlooked the role of global monetary conditions in shaping equity market fluctuations. Second, unlike prior studies that examine isolated dimensions of market behaviour, this paper offers a more comprehensive characterisation of macro-financial transmission by jointly analysing Granger causality, impulse response functions, and forecast error variance decomposition within a single empirical framework – allowing for a more internally consistent assessment of both the direction and magnitude of shock transmission to equity returns. Third, this paper contributes to the broader debate on the informational efficiency of emerging equity markets by assessing the extent to which observable macroeconomic fundamentals can account for VN-Index fluctuations, and by quantifying the role of idiosyncratic, non-fundamental factors in driving equity dynamics in a frontier market setting.

The remainder of this paper is organised as follows. Section II reviews the theoretical background and empirical literature on macro-financial dynamics and stock market behaviour within a VAR framework. Section III introduces the data, variable construction, and empirical methodology employed in the analysis. Section IV presents the main empirical findings, including VAR estimation results, impulse response analysis, and variance decomposition. Section V discusses the broader economic implications of the findings for investors and policymakers. Finally, Section VI concludes the paper.

II. LITERATURE REVIEW

A *Theoretical Background: APT and Macroeconomic Risk Factors*

The theoretical foundation for a macroeconomic approach to equity pricing rests on the Arbitrage Pricing Theory (APT) proposed by [1], which posits that expected equity returns are determined by multiple systematic risk factors rather than the single market factor of the Capital Asset Pricing Model [10]. Under APT, any variable that generates pervasive, undiversifiable risk must be compensated through expected returns; the key empirical challenge is to identify which macroeconomic variables constitute such systematic factors.

[2] provide the most influential empirical operationalisation of APT, identifying industrial production, unexpected inflation, the term structure of interest rates, and the spread between high- and low-grade bonds as systematically priced sources of macroeconomic risk in the U.S. equity market over 1953–1983. These variables represent the principal channels through which the real economy transmits information to equity valuations: changes in interest rates alter the discount rate applied to future cash flows; inflation erodes the real value of nominal returns; and industrial production proxies for aggregate economic activity and expected cor-

porate earnings. The [2] framework motivates the inclusion of CPI, deposit rate, NEER, oil price, and FFR as candidate macroeconomic shocks and their transmission to VN-Index returns in the present study, with each variable representing a distinct channel of systematic risk transmission.

The efficient market hypothesis [11] provides a complementary theoretical perspective: under EMH, stock prices fully and instantaneously reflect all publicly available information, including macroeconomic announcements. A direct implication is that only unexpected components of macroeconomic changes should generate equity price responses, since expected changes are already embedded in current valuations. [4] demonstrates that the negative stock-return–inflation relationship documented in the empirical literature is a spurious reflection of the positive correlation between real activity and equity returns: anticipated inflation is priced in, while unexpected inflation generates real return shocks. This distinction between anticipated and unanticipated macroeconomic changes underlies the impulse response methodology adopted in the present paper.

B *Monetary Policy and Equity Markets: Evidence from Developed Markets*

The relationship between monetary policy and equity market returns has attracted substantial empirical attention, particularly in the context of U.S. monetary policy shocks. [3] provide a rigorous decomposition of how Federal Reserve policy affects U.S. equity prices by separating anticipated and unanticipated components using Federal funds futures contracts. Their key finding is that equity markets respond primarily to the surprise component of policy actions, with an unexpected 25-basis-point reduction in the FFR increasing broad equity indices by approximately 1%, while anticipated changes have negligible effects.

More importantly for emerging market contexts such as Vietnam, their VAR-based decomposition

highlights that the transmission of monetary policy operates primarily through the equity risk premium channel. In this framework, unexpected monetary tightening increases the required return on risky assets, leading to equity price declines even when real interest rate effects are limited. In small open economies, this mechanism is likely to operate not only through domestic financial conditions but also through international capital flow adjustments and global investor risk rebalancing, which are central transmission channels for the VN-Index.

Complementary evidence from the international monetary spillover literature reinforces this interpretation. [8] show that U.S. monetary tightening generates significant capital outflow pressures in emerging markets through portfolio rebalancing and risk appetite channels, with stronger effects during rapid hiking cycles. This mechanism is particularly relevant for the 2022–2023 Federal Reserve tightening episode included in the present study, suggesting that FFR shocks can be expected to exert pronounced effects on Vietnamese equity market dynamics through global liquidity and capital flow channels.

C Macroeconomic Shocks and Stock Market Dynamics in Emerging Markets

The transmission of macroeconomic shocks to equity markets in emerging economies differs systematically from that in developed markets due to differences in financial depth, investor composition, liquidity conditions, and institutional frameworks. [7] show that international economic shocks are incorporated into equity prices even under partial market segmentation, with the degree of transmission depending on the openness of capital accounts and financial integration with global markets. This suggests that external shocks may play a relatively more important role in driving equity dynamics in emerging markets compared to purely domestic macroeconomic innovations.

More recent literature highlights the impor-

tance of the global financial cycle in amplifying these transmission mechanisms. [12] argues that global monetary conditions, particularly those driven by U.S. monetary policy, generate synchronised movements in capital flows, credit conditions, and risky asset prices across countries, even under flexible exchange-rate regimes. This mechanism is particularly relevant for small open economies such as Vietnam, where equity markets are highly sensitive to changes in global liquidity and investor risk appetite. Accordingly, U.S. monetary policy shocks are expected to represent a key external driver of VN-Index fluctuations through capital flow and portfolio reallocation channels.

In addition to financial shocks, global supply-side factors also contribute to macro-financial volatility in emerging markets. [6] documents that oil price shocks significantly affect equity returns through their impact on production costs and expected corporate profitability. In oil-importing economies, these effects are further amplified through inflationary pressures and exchange rate adjustments, making oil prices an important external determinant of equity market dynamics in emerging markets.

D Prior Evidence on the Vietnamese Stock Market

Empirical research on the macroeconomic determinants of VN-Index returns remains relatively scarce and fragmented compared to the broader emerging market literature. Existing studies primarily focus on specific transmission channels rather than a comprehensive macro-financial system.

[9] analyse monetary policy transmission in Vietnam using a VAR framework over the period 2003–2012, focusing on interest rate, exchange rate, and asset price channels. Their results indicate a one-directional relationship from CPI to VN-Index, suggesting that inflation dynamics are reflected in equity market movements, while feedback effects

from the stock market to CPI are limited. They also document evidence of a cost channel in which increases in interbank interest rates (VNIBOR) raise production costs and inflation, rather than reducing economic activity through conventional demand-side mechanisms. The weak effectiveness of the exchange rate channel is attributed to the State Bank of Vietnam’s administrative management of the USD/VND rate.

[5] extend this analysis using a time-varying structural VAR framework over the period 2000–2016, incorporating both domestic macroeconomic conditions and regional market linkages. Their findings suggest that domestic macroeconomic factors such as monetary easing, currency stability, and economic growth positively affect stock market development. In addition, regional equity markets (Thailand, Japan, Hong Kong, and China) play a significant role in driving VN-Index fluctuations. Importantly, they document a decline in the sensitivity of VN-Index to external shocks over time, suggesting gradual market maturation and improved price discovery.

Taken together, these studies suggest that while domestic macroeconomic fundamentals remain relevant, external financial shocks, particularly global commodity prices and U.S. monetary policy conditions, are increasingly important in explaining Vietnamese equity market dynamics. However, existing evidence remains fragmented, as studies typically focus either on domestic transmission channels or regional integration, without jointly modelling domestic and external macroeconomic shocks within a unified framework. This motivates a VAR-based approach that incorporates both domestic variables (CPI, deposit rate, exchange rate) and external shocks (oil prices, U.S. monetary policy) to capture the full macro-financial transmission mechanism affecting the VN-Index.

E Research Gaps

Despite the growing literature on macro-financial dynamics in Vietnamese equity markets, several important gaps remain. First, much of the existing empirical evidence does not cover the period after 2020, during which Vietnamese equity markets were exposed to significant external shocks – most notably the rapid tightening of U.S. monetary policy during 2022–2023. As a result, how the VN-Index responds to heightened global financial volatility in more recent years remains underexplored. Second, the role of global monetary policy shocks, particularly U.S. interest rate dynamics, has not been fully incorporated into empirical VAR-based analyses of Vietnamese equity markets alongside domestic macroeconomic variables. This limits understanding of how external financial cycles transmit to domestic equity market dynamics in an open emerging economy context. Third, existing empirical approaches are often fragmented, focusing separately on Granger causality, dynamic shock responses, or variance decomposition, rather than integrating these elements within a unified VAR framework. This restricts a comprehensive evaluation of both the direction and magnitude of macroeconomic shock transmission to Vietnamese equity returns.

III. DATA AND METHODOLOGY

A Data

This study employs monthly data spanning January 2010 to December 2024, yielding 180 observations. The sample period is chosen to capture a sufficiently long horizon that encompasses both stable and turbulent phases of Vietnamese equity market development, including the Federal Reserve’s 2022–2023 rate-hiking cycle – the most aggressive U.S. monetary tightening in four decades – which is of particular relevance given the paper’s focus on external monetary policy transmission.

Variable selection is guided by the Arbitrage

Pricing Theory of [2], under which macroeconomic variables represent systematic risk factors that generate undiversifiable risk and are therefore priced into equity returns. Six variables are included, classified by their economic role and described in Table 1. The VN-Index (VNI) serves as the dependent variable, representing the benchmark equity return of the Ho Chi Minh Stock Exchange. Consumer price inflation (CPI) is included because unexpected inflation erodes the real value of corporate cash flows and raises uncertainty over future earnings [2, 4]. The deposit rate (DEP) is included as a proxy for domestic monetary conditions and the opportunity cost of equity investment. An increase in deposit rates raises the relative attractiveness of risk-free savings instruments compared with equities, thereby reducing investor demand for stocks and placing downward pressure on VN-Index valuations. This channel is particularly relevant in the Vietnamese equity market, where retail investors account for the majority of trading activity and are highly sensitive to changes in deposit returns. The nominal effective exchange rate (NEER) captures the competitiveness channel: exchange rate movements alter the relative profitability of internationally exposed firms and influence investor sentiment toward the domestic currency [5, 7]. The global oil price (OIL) represents supply-side cost shocks that compress corporate profit margins and reduce expected dividends [6]. Finally, the U.S. Federal Funds Rate (FFR) is included as the key external monetary policy variable. [7] demonstrate that international economic forces are priced into equity markets even under market segmentation, while [3] show that unexpected FFR changes shift the global equity risk premium through capital flow and risk appetite channels— effects that were particularly pronounced during the 2022–2023 tightening cycle [8].

Table 1: Variable Descriptions and Sources

Variable	Abbrev.	Symbol	Transform
VN-Index	VNI	VNI_t	$\ln(VNI_t)$
CPI YoY	CPI	CPI_t	Level (%)
Deposit Rate	DEP	DEP_t	Level (%)
Nominal Eff. Exchange Rate	NEER	$NEER_t$	$\ln(NEER_t)$
Fed Funds Rate	FFR	FFR_t	Level (%)
Oil Price (Brent)	OIL	OIL_t	$\ln(OIL_t)$

The natural logarithm is applied to VNI, NEER, and OIL to stabilise variance and ensure that subsequent first differences yield interpretable percentage changes; CPI, DEP, and FFR are retained in level form as they are already expressed in percentage terms. Whether first differencing is required for these variables is determined empirically in Section B.

Descriptive statistics for the transformed variables in levels are reported in Table 2. CPI exhibits pronounced right-skewness (skewness = 2.18), driven by the inflation spike of 2011 (peak: 23.0%), followed by a sustained disinflation trend to below 5% from 2013 onward. FFR displays similarly high skewness (1.48), reflecting the decade-long near-zero rate environment of 2010–2021 and the abrupt hiking cycle of 2022–2023 during which the Federal Funds rate rose from 0.05% to 5.33%. These distributional features motivate the use of log transformation and first differencing, the necessity of which is formally established in the methodology section.

B Methodology

The empirical procedure follows the standard VAR framework of [13], proceeding through stationarity testing, cointegration analysis, lag selection, model estimation and adequacy checking, and dynamic analysis. The VAR treats all variables as jointly endogenous and imposes no a priori re-

Table 2: Descriptive Statistics – Transformed Variables (2010M01–2024M12)

Stat	ln(VNI)	CPI (%)	DEP (%)	ln(NEER)	FFR (%)	ln(OIL)
<i>N</i>	180	180	180	180	180	180
Mean	6.62	5.07	8.51	10.00	1.23	4.23
Std	0.41	4.66	2.32	0.07	1.72	0.33
Min	5.86	−0.98	5.90	9.80	0.05	2.81
p25	6.23	2.55	7.20	9.95	0.09	3.99
Median	6.62	3.50	7.30	10.02	0.19	4.29
p75	6.97	5.95	9.00	10.05	1.82	4.49
Max	7.31	23.02	14.00	10.10	5.33	4.74

strictions on the structural relationships between variables. Because the VAR expresses each dependent variable solely in terms of predetermined lagged values, it constitutes a reduced-form model that is well suited to characterising the dynamic interactions among macroeconomic variables and equity returns.

1. Stationarity Testing

The VAR framework requires all variables entering the system to be stationary $I(0)$. A clearly important argument that arises when dealing with stock market data is whether to use levels or first differences in the VAR. The majority view, as highlighted by [14] and [15], is that stationary data should be used, since non-stationary data can lead to spurious regression results. Accordingly, stationarity is assessed using three complementary procedures with opposing null hypotheses to reduce the risk of reliance on any single test.

Augmented Dickey-Fuller (ADF) test. The ADF regression with constant is:

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t, \quad (1)$$

$$H_0 : \gamma = 0 \quad (\text{unit root})$$

The lag order k is selected by AIC with a maximum of 12. Rejection of H_0 at the 5% level provides evidence of stationarity.

Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. The KPSS procedure reverses the null: H_0 states that the series *is* stationary, based on a Lagrange Multiplier score testing principle [16]. A variable is confirmed $I(0)$ only when ADF rejects H_0 and KPSS fails to reject H_0 , thereby reducing the risk of spurious conclusions from either test alone.

Phillips–Perron (PP) test. Where ADF and KPSS yield conflicting or inconclusive results, the PP test is applied. This test uses a non-parametric correction for serial correlation and heteroskedasticity in the regression residuals [17], making it more robust than ADF in the presence of structural breaks or time-varying variance.

Results. Applied to the level series, all six variables fail the joint ADF/KPSS stationarity condition – ADF fails to reject the unit root null and KPSS rejects the stationarity null – confirming that all variables are $I(1)$. The Johansen cointegration test is therefore applied before first differencing to determine the appropriate estimation framework, as described in the following subsection.

2. Johansen Cointegration Test

Since all six variables are confirmed $I(1)$ in levels, the Johansen procedure is applied to test whether a long-run cointegrating relationship exists among them. If cointegration is detected, a Vector Error Correction Model (VECM) is required to

preserve long-run information; absent cointegration, a VAR on first differences is appropriate. It has been argued that, even in the presence of cointegration, unrestricted VARs perform well at short horizons [18, 19], and [20] demonstrates that the impulse response results from unrestricted VARs and VECMs are nearly identical in short-run analysis. Nevertheless, the cointegration test is carried out as a formal specification check.

The Johansen procedure tests the rank r of the long-run multiplier matrix sequentially using two complementary statistics. Both the Trace statistic ($84.22 < 95.75$ at the 5% level) and the Maximum-Eigenvalue statistic ($33.39 < 40.08$ at the 5% level) fail to reject the null of zero cointegrating vectors at the first step of the sequential procedure, indicating the absence of a long-run equilibrium relationship among the six variables. A VAR on first differences is therefore appropriate, and a VECM is not required.

Given these results, first differences are applied to all variables. To confirm that first differencing achieves stationarity, ADF and KPSS tests are re-applied to all differenced series. Five variables are straightforwardly confirmed $I(0)$, with ADF p -values ranging from 0.000 to 0.005 and KPSS statistics falling uniformly below the 5% critical value. For ΔFFR , the ADF test yields an inconclusive result ($p = 0.096$), attributable to the structural break between the near-zero rate regime of 2010–2021 and the aggressive hiking cycle of 2022–2023, which reduces the test's power to distinguish a unit root from a stationary process with a level shift. The PP test resolves this ambiguity by rejecting the unit root null ($p = 0.000$), confirming that ΔFFR is $I(0)$. All six first-differenced variables are therefore confirmed stationary and enter the system vector:

$$\mathbf{y}_t = \begin{pmatrix} \Delta\text{FFR}_t, \Delta \ln \text{OIL}_t, \Delta \text{CPI}_t, \\ \Delta \text{DEP}_t, \Delta \ln \text{NEER}_t, \Delta \ln \text{VNI}_t \end{pmatrix}' \quad (2)$$

where $\Delta x_t = x_t - x_{t-1}$ denotes the first difference

and $\Delta \ln x_t \approx$ the percentage change in x_t .

3. Optimal Lag Length Selection

The appropriate number of lags is an important specification decision in any VAR analysis. To determine the optimal lag length p , both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are computed for VAR models with lag orders ranging from $p = 0, 1, \dots, 12$:

$$\text{AIC}(p) = \ln|\hat{\Sigma}_u(p)| + \frac{2}{T}pk^2 \quad (3)$$

$$\text{BIC}(p) = \ln|\hat{\Sigma}_u(p)| + \frac{\ln T}{T}pk^2 \quad (4)$$

where $\hat{\Sigma}_u(p)$ denotes the estimated residual covariance matrix at lag order p , $k = 6$ is the number of endogenous variables, and $T = 177$ is the effective sample size after differencing and lag adjustment.

BIC is prioritised because it applies a stronger penalty for adding extra lags, which generally leads to simpler and more stable models in relatively small samples. BIC selects $p_0 = 1$, while AIC selects $p_0 = 2$. To assess whether one lag is sufficient, the multivariate Portmanteau test is applied to the residuals of the VAR(1) model. The null hypothesis of no residual autocorrelation is rejected ($p = 0.021 < 0.05$), implying that one lag is insufficient to capture the dynamic structure of the system. Increasing the lag length to $p_1 = 2$ eliminates this problem: the Portmanteau test fails to reject the null ($p = 0.143 > 0.05$), confirming that VAR(2) residuals behave as white noise. The final specification adopted throughout this study is therefore VAR(2).

4. VAR Specification

The VAR treats all variables as jointly endogenous and imposes no a priori restrictions on the structural relationships between variables. This symmetric treatment follows [13], who argues that imposing incredible identification restrictions on structural relationships between variables is unnec-

essary when the goal is empirical characterisation of dynamic interactions.

The reduced-form VAR(2) model estimated on first-differenced variables is:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + u_t, \quad (5)$$

where $u_t \sim (0, \Sigma_u)$

where c is a (6×1) intercept vector; A_1 and A_2 are (6×6) matrices of autoregressive coefficients; and u_t is the vector of reduced-form residuals with zero mean and covariance matrix Σ_u . Each equation is estimated by OLS, which yields consistent and asymptotically efficient estimates under stability and white-noise residuals.

The VN-Index equation of primary interest is:

$$\begin{aligned} \Delta \ln \text{VNI}_t = & c_1 + a_{11}^{(1)} \Delta \ln \text{VNI}_{t-1} + a_{12}^{(1)} \Delta \text{CPI}_{t-1} \\ & + a_{13}^{(1)} \Delta \text{DEP}_{t-1} + a_{14}^{(1)} \Delta \ln \text{NEER}_{t-1} \\ & + a_{15}^{(1)} \Delta \text{FFR}_{t-1} + a_{16}^{(1)} \Delta \ln \text{OIL}_{t-1} \\ & + a_{11}^{(2)} \Delta \ln \text{VNI}_{t-2} + a_{12}^{(2)} \Delta \text{CPI}_{t-2} \\ & + a_{13}^{(2)} \Delta \text{DEP}_{t-2} + a_{14}^{(2)} \Delta \ln \text{NEER}_{t-2} \\ & + a_{15}^{(2)} \Delta \text{FFR}_{t-2} + a_{16}^{(2)} \Delta \ln \text{OIL}_{t-2} + u_{1t} \end{aligned} \quad (6)$$

The five remaining equations follow an identical structure. Of particular interest is $\hat{a}_{15}^{(1)} = -0.0996$ ($p = 0.025$), the estimated one-lag effect of FFR changes on VN-Index returns, which is examined further in the Granger causality and impulse response analyses.

OLS Estimation. Writing equation (5) in stacked matrix form across all T time periods:

$$Y = XB + E \quad (7)$$

where:

$$\begin{aligned} Y &= (y_{p+1}, \dots, y_T)' \in \mathbb{R}^{T \times k} \\ X &= (x_{p+1}, \dots, x_T)' \in \mathbb{R}^{T \times m} \\ x_t &= (1, y'_{t-1}, y'_{t-2})' \\ B &= (c, A_1, A_2)' \in \mathbb{R}^{m \times k} \end{aligned}$$

$$E = (u_{p+1}, \dots, u_T)'$$

with $k = 6$ endogenous variables, $p = 2$ lags, and $m = 1 + kp = 13$ regressors per equation. The OLS estimator applied equation-by-equation is:

$$\hat{B} = (X'X)^{-1}X'Y \quad (8)$$

Because all equations share the same regressor matrix X , equation-by-equation OLS estimation yields efficient estimates and is sufficient for estimating the VAR system [21]. The estimator $\hat{B}$ is consistent and asymptotically normal under the conditions of model stability and white-noise residuals, which are verified through the diagnostic checks.

Residual Covariance Estimation. The residual covariance matrix Σ_u is estimated from the OLS residuals as:

$$\hat{\Sigma}_u = \frac{1}{T - m} \hat{E}'\hat{E} \quad (9)$$

where $\hat{E} = Y - X\hat{B}$ is the $(T \times k)$ matrix of OLS residuals, $T = 177$ is the effective sample size, and $m = 1 + kp = 13$ is the number of regressors per equation. The degrees-of-freedom correction $T - m$ yields an unbiased estimate of Σ_u [21]. The off-diagonal elements of $\hat{\Sigma}_u$ measure contemporaneous shock correlations across equations. Non-negligible correlations motivate Cholesky orthogonalisation for structural identification in the impulse response analysis, as described in Section 7.

5. ARDL Model as Robustness Check

To assess the robustness of the VAR-based findings, this study estimates an Autoregressive Distributed Lag (ARDL) specification as a single-equation dynamic alternative. Unlike the VAR, which treats all variables as jointly endogenous within a system of equations, the ARDL isolates the VN-Index equation and allows flexible lag distributions across both the dependent variable and

each macroeconomic regressor.

Given that the Johansen procedure detects no cointegrating relationship among the six variables, the ARDL model is estimated on first-differenced variables, consistent with the stationarity conditions established in Section B. Following [22], the specification takes the form:

$$\begin{aligned} \Delta \ln \text{VNI}_t = & \alpha + \sum_{i=1}^p \phi_i \Delta \ln \text{VNI}_{t-i} + \sum_{j=0}^{q_1} \beta_{1j} \Delta \text{FFR}_{t-j} \\ & + \sum_{j=0}^{q_2} \beta_{2j} \Delta \ln \text{OIL}_{t-j} + \sum_{j=0}^{q_3} \beta_{3j} \Delta \text{CPI}_{t-j} \\ & + \sum_{j=0}^{q_4} \beta_{4j} \Delta \text{DEP}_{t-j} + \sum_{j=0}^{q_5} \beta_{5j} \Delta \ln \text{NEER}_{t-j} + \varepsilon_t \end{aligned} \quad (10)$$

The lag structure $(p, q_1, \dots, q_5) = (2, 2, 2, 2, 2, 2)$ mirrors the VAR(2) specification, ensuring comparability of short-run dynamic responses across the two frameworks. Heteroskedasticity-robust standard errors (HC3) are employed throughout, consistent with the ARCH effects detected in some equations of the VAR system.

Since no cointegration is detected, the bounds testing procedure of [22] is not applicable, and the model is not interpreted within an error-correction or long-run equilibrium framework. The ARDL specification serves strictly as a consistency check on the sign, magnitude, and significance of the short-run macro-financial transmission coefficients identified in the VAR analysis, with particular attention to the FFR channel.

6. Model Adequacy Checks

Several post-estimation diagnostic tests are conducted to verify that the VAR(2) specification satisfies the assumptions required for valid impulse response and variance decomposition analysis, with results summarised in Table 3. The Portmanteau test is evaluated at lag 12, consistent with the monthly frequency of the data and the convention of testing up to one full seasonal cy-

cle [21]. The ARCH-LM test uses $q = 5$ lags, sufficient to detect short-run volatility clustering typical of financial time series.

Table 3: Model Adequacy Diagnostics – VAR(2)

Test	Null Hypothesis	Result	Conclusion
Stability	All roots inside unit circle	Max = 0.756	Pass
Portmanteau (lag 12)	No serial correlation in residuals	$p = 0.143$	Pass
Jarque–Bera	Residuals are normally distributed	$p = 0.000$	Fail ^a
ARCH-LM ($q = 5$)	Constant variance in residuals	DEP, OIL fail	Noted ^b
Residual correlation	Shocks are uncorrelated across equations	FFR–OIL: $r = 0.44$	Cholesky applied

^a Residuals are non-normal due to the extreme COVID-19 market shock of March 2020.

^b ARCH effects are detected in the DEP and OIL equations only.

The Jarque–Bera test rejects residual normality, primarily due to extreme outlier observations generated by the COVID-19 market crash of March 2020 – an episode that falls within the sample period 2010–2024. Although OLS estimates remain consistent under non-normal residuals [23], the asymptotic confidence bands for impulse responses should nevertheless be interpreted with caution.

ARCH-LM tests further detect time-varying residual variance in the DEP and OIL equations, consistent with the volatility clustering commonly observed in interest rate and commodity markets [24]. By contrast, the VNI equation itself shows no evidence of ARCH effects ($p = 0.992$), suggesting that the main impulse response results for VN-Index remain reasonably reliable. However, residual heteroskedasticity may still cause asymptotic IRF confidence bands to understate true uncertainty during highly volatile periods [21].

Finally, the residual correlation matrix reveals several non-negligible contemporaneous correlations, most notably between FFR and OIL ($r = 0.439$), as well as between VNI and OIL ($r = 0.301$) and between VNI and FFR ($r = 0.244$). These results confirm that reduced-form shocks

are correlated across equations within the same period, thereby motivating the use of Cholesky orthogonalisation for structural identification in the impulse response analysis discussed in Section 7.

7. Dynamic Analysis

Granger Causality. For each predictor x_{jt} , the pairwise Granger causality test evaluates whether past values of x_j improve the prediction of $\Delta \ln \text{VNI}_t$ beyond its own history. The null hypothesis is:

$$H_0 : a_{1j}^{(1)} = a_{1j}^{(2)} = 0 \quad (11)$$

tested via an F -statistic at the two-lag horizon consistent with the VAR(2) specification. Rejection at the 5% level indicates that x_j Granger-causes $\Delta \ln \text{VNI}_t$. As emphasised by [25], this reflects predictive precedence rather than structural causality.

Orthogonalized Impulse Response Functions (OIRF). Because the reduced-form residuals are contemporaneously correlated ($\hat{\Sigma}_u \neq I$), structural identification is achieved via Cholesky decomposition:

$$\hat{\Sigma}_u = PP', \quad P \text{ is lower triangular} \quad (12)$$

The OIRF of VN-Index at horizon h to a unit structural shock in variable j is:

$$\text{OIRF}(h, j) = \Phi_h p_j \quad (13)$$

where Φ_h is the h -step moving average matrix and p_j is the j -th column of P . The Cholesky ordering adopted is:

$$\begin{aligned} \Delta \text{FFR}_t &\rightarrow \Delta \ln \text{OIL}_t \rightarrow \Delta \text{CPI}_t \rightarrow \\ \Delta \text{DEP}_t &\rightarrow \Delta \ln \text{NEER}_t \rightarrow \Delta \ln \text{VNI}_t \end{aligned} \quad (14)$$

This ordering places global variables first on the grounds that U.S. monetary policy and international commodity prices are contemporaneously exogenous to the Vietnamese economy within a

given month [7]. VN-Index is ordered last because financial asset prices are assumed to adjust instantaneously to all available information, a standard assumption in the asset pricing literature [3]. Confidence bands are computed at the 90% level using asymptotic standard errors over a 24-month horizon.

Forecast Error Variance Decomposition (FEVD). The FEVD quantifies the percentage of the h -step VN-Index forecast error variance attributable to each structural shock:

$$\omega_j(h) = \frac{\sum_{s=0}^{h-1} (\Phi_s p_j)_1^2}{\text{MSE}_h(\Delta \ln \text{VNI}_t)} \times 100\%, \quad (15)$$

where $\sum_{j=1}^6 \omega_j(h) = 100\%$

FEVD is computed at horizons $h \in \{1, 3, 6, 12, 18\}$ months to capture both short-run and medium-run dynamics.

IV. EMPIRICAL RESULTS

A ARDL Robustness Check

To verify the robustness of the VAR-based coefficient estimates, an ARDL(2, 2, 2, 2, 2, 2) model is estimated for the $\Delta \ln(\text{VNI})$ equation using the same set of first-differenced variables. Lagged coefficients (L1 and L2) from both specifications are reported in Table 4; contemporaneous terms (L0) present in the ARDL but absent from the VAR are excluded to maintain comparability.

Table 4: VAR(2) vs. ARDL(2, 2, 2, 2, 2, 2):
Short-run Coefficient Comparison, $\Delta \ln(\text{VNI})$
Equation

Variable	VAR(2)		ARDL	
	Coef.	p	Coef.	p
ΔFFR_{t-1}	-0.0996	0.025	-0.1586	0.154
ΔFFR_{t-2}	+0.0349	0.442	+0.0522	0.436
ΔCPI_{t-1}	-0.0025	0.741	-0.0002	0.990
ΔCPI_{t-2}	-0.0007	0.929	+0.0012	0.890
ΔDEP_{t-1}	-0.0097	0.569	-0.0107	0.521
ΔDEP_{t-2}	+0.0009	0.959	+0.0187	0.347
$\Delta \ln \text{NEER}_{t-1}$	-0.2199	0.730	-0.4690	0.726
$\Delta \ln \text{NEER}_{t-2}$	+0.3626	0.558	+0.6864	0.337
$\Delta \ln \text{OIL}_{t-1}$	-0.0200	0.713	-0.0363	0.494
$\Delta \ln \text{OIL}_{t-2}$	+0.0164	0.730	+0.0313	0.580

Notes: All variables in first differences. NEER and OIL in log-differenced form. ARDL p -values based on HC3 robust standard errors.

Sign consistency is confirmed for nine of ten lagged coefficients. The ΔFFR_{t-1} coefficient is negative in both specifications (-0.100 in VAR, -0.159 in ARDL), although individual significance is attained only in the VAR.

The loss of significance in the ARDL specification is attributable to two compounding factors. First, HC3 robust standard errors impose a more conservative correction for heteroskedasticity than the asymptotic standard errors used in the VAR. Second, the single-equation ARDL foregoes the efficiency gains from joint system estimation, resulting in larger standard errors for individual coefficients. Notably, the point estimate increases in magnitude from -0.100 to -0.159 across specifications, confirming that the direction of the FFR effect is consistent and that the VAR result is not an artefact of system-based estimation.

All remaining variables display consistent signs and comparable magnitudes across both models. The sole exception – ΔCPI_{t-2} – involves coefficients that are economically negligible and statistically insignificant in both specifications. Overall,

the cross-specification consistency reduces concerns regarding model dependence and supports the credibility of the VAR-based findings reported above.

B Granger Causality

Table 5 reports the pairwise Granger causality test results for the null hypothesis that each macroeconomic variable does not Granger-cause $\Delta \ln \text{VNI}_t$, evaluated at the two-lag horizon consistent with the VAR(2) specification.

The results reveal that ΔFFR is the only variable that Granger-causes $\Delta \ln \text{VNI}_t$ at the 5% significance level ($F = 4.395$, $p = 0.014$). The joint F -test on both lags confirms that past changes in the FFR contain incremental predictive information for VN-Index returns beyond the index's own history. In contrast, none of the domestic variables – CPI, Deposit Rate, or NEER – nor the global oil price exhibits statistically significant predictive power at conventional significance levels.

This finding is consistent with [3], who argue that U.S. monetary policy affects equity markets primarily through the equity risk premium channel. A tightening of the Federal Funds Rate increases the required return on risky assets by raising perceived financial risk and reducing investors' willingness to bear risk, thereby lowering equity valuations. For emerging markets such as Vietnam, tighter U.S. monetary policy may also weaken global risk appetite and reduce capital flows toward riskier emerging-market assets, placing downward pressure on the VN-Index.

The absence of Granger causality from domestic variables does not necessarily imply that these variables have no effect on the VN-Index. As noted by [5], emerging market equity prices tend to respond to macroeconomic shocks contemporaneously rather than with a lag, so their influence may be captured in the impulse response analysis rather than in the Granger causality test.

Table 5: Pairwise Granger Causality Tests: $X \nleftrightarrow \Delta \ln \text{VNI}$

Predictor (X)	F (lag 1)	p (lag 1)	F (lag 1&2)	p (lag 1&2)	Decision
ΔFFR	7.938	0.005	4.395	0.014	Granger-causes VNI **
$\Delta \ln \text{OIL}$	1.922	0.167	0.895	0.411	No evidence
ΔCPI	0.577	0.449	0.346	0.708	No evidence
ΔDEP	0.789	0.376	0.426	0.654	No evidence
$\Delta \ln \text{NEER}$	0.281	0.597	0.179	0.836	No evidence

Note: F (lag 1) is the F -statistic for the first lag only; F (lag 1&2) is the joint test on both lags, consistent with the VAR(2) specification. Significance: ** $p < 0.05$. H_0 : X does not Granger-cause $\Delta \ln \text{VNI}_t$.

C Impulse Response Analysis

Figure 1 presents the orthogonalized impulse response functions (OIRFs) of $\Delta \ln \text{VNI}_t$ to a one-standard-deviation structural shock in each macroeconomic variable over a 24-month horizon. To provide a precise numerical reference for the short-run dynamics, Table 6 summarizes the impact and early-stage responses for the first three months (Month 0 to Month 2).

Response to FFR shock. As detailed in Table 6, a one-standard-deviation increase in the policy rate (FFR) generates an immediate positive response in VN-Index returns of approximately 0.0145 on impact (Month 0). One possible explanation is that the contemporaneous response reflects short-run adjustment dynamics, anticipatory pricing effects, or temporary domestic policy conditions that partially offset the external tightening shock during the initial period. However, the response reverses sharply to -0.0117 in Month 1 before gradually converging toward zero, suggesting that the adverse effects of tighter global monetary conditions emerge more clearly with a lag. This delayed negative adjustment is broadly consistent with transmission through capital outflow pressures, tighter global liquidity conditions, and changes in investor risk appetite. The cumulative response becomes negative after the second month and stabilizes at approximately -0.011 over the 24-month horizon.

Response to Oil Price shock. An increase in oil prices produces an immediate positive response

in VN-Index returns of approximately 0.0128 on impact. The response declines rapidly toward zero in subsequent periods. The cumulative response remains modest, stabilizing at approximately 0.013 over the 24-month horizon.

Response to CPI shock. A CPI shock generates a small negative response in VN-Index returns of approximately -0.0074 on impact. The effect weakens quickly and converges toward zero within several periods. The cumulative response remains negative throughout the forecast horizon, reaching approximately -0.016 after 24 months.

Response to Deposit Rate shock. A positive shock to the deposit rate leads to a small positive impact response of approximately 0.0072. The response subsequently turns slightly negative before converging toward zero. Overall, the effect remains economically small, with the cumulative response stabilizing at approximately 0.002 over the 24-month horizon.

Response to NEER shock. A depreciation shock (increase in NEER) generates an immediate negative response in VN-Index returns of approximately -0.0128 . The response gradually weakens and approaches zero after several periods. The cumulative response converges to approximately -0.015 after 24 months.

Summary. Overall, the IRF results indicate that VN-Index responses to macroeconomic shocks are generally short-lived and economically modest. A notable feature across several shocks – particu-

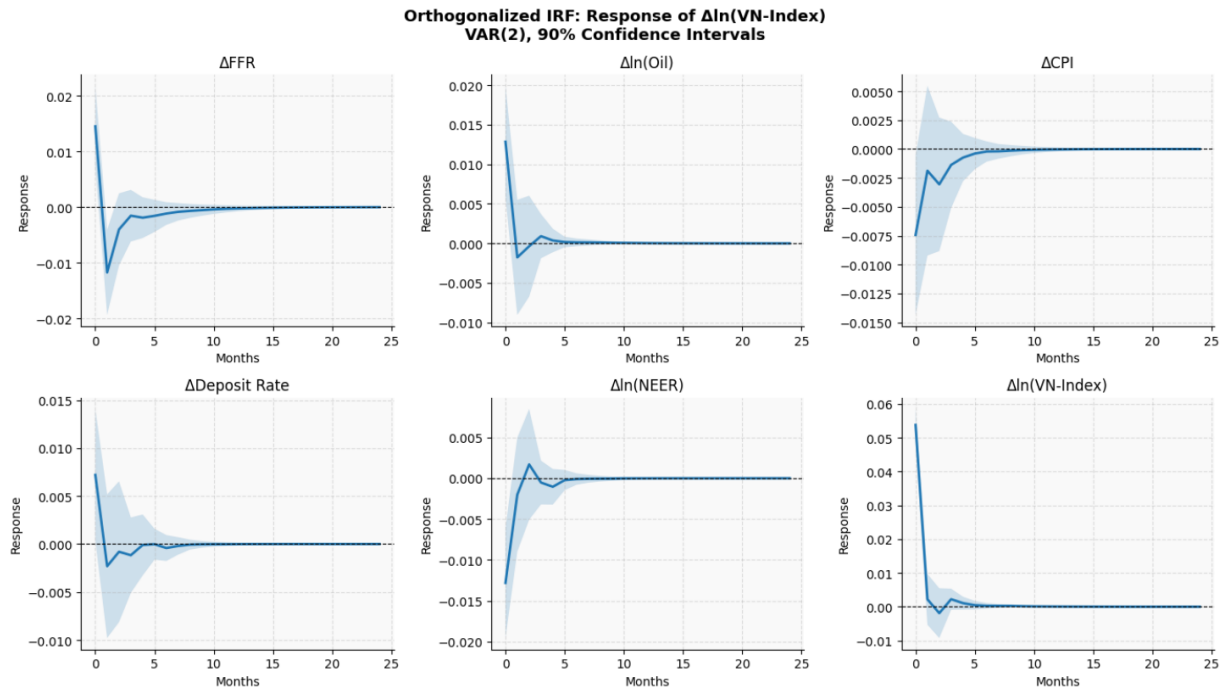


Figure 1: Orthogonalized Impulse Responses of $\Delta \ln \text{VNI}_t$ to Macroeconomic Shocks.

Note: Shaded bands = 90% asymptotic CI.

larly FFR, oil price, and deposit rate shocks – is the presence of small positive impact-period responses followed by rapid reversals toward zero or negative territory. One possible explanation is that the transmission of macroeconomic shocks to Vietnamese equity prices may occur with short lags, reflecting gradual market adjustment dynamics and the interaction between external shocks and domestic financial conditions. In the case of oil prices, the temporary positive response may also capture offsetting effects associated with Vietnam’s exposure to both commodity-exporting and commodity-importing activities. Most responses converge rapidly toward zero within several periods, and the majority of responses fall within the 90% confidence bands beyond the short run. The cumulative impulse responses further suggest that macroeconomic shocks exert only limited net effects on Vietnamese equity returns over the 24-month horizon.

Table 6: Summary of Orthogonalized Impulse Responses of $\Delta \ln \text{VNI}_t$

Shock	Month 0	Month 1	Month 2
FFR	0.0145	-0.0117	-0.0040
Oil Price	0.0128	-0.0018	-0.00036
CPI	-0.0074	-0.0019	-0.0031
Deposit Rate	0.0072	-0.0023	-0.0008
NEER	-0.0128	-0.0020	0.0017

D Forecast Error Variance Decomposition

Table 7 presents the FEVD of $\Delta \ln \text{VNI}_t$ at selected horizons from 1 to 18 months. The decomposition quantifies the percentage of VN-Index forecast error variance attributable to each structural shock, with each row summing to 100%. Three main findings emerge from the FEVD results.

First, own shocks account for the dominant share of VN-Index forecast error variance at all horizons, declining only slightly from 78.3% at $h = 1$ to approximately 77.4% at longer horizons. This suggests that a substantial portion of VN-

Table 7: Forecast Error Variance Decomposition of $\Delta \ln \text{VNI}_t$ (%)

Shock Variable	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 18$
$\Delta \ln \text{VNI}$	78.34	77.64	77.45	77.41	77.41
ΔFFR	9.42	9.80	9.97	10.01	10.01
$\Delta \ln \text{NEER}$	4.55	4.58	4.60	4.59	4.59
$\Delta \ln \text{OIL}$	4.55	4.52	4.51	4.51	4.51
ΔCPI	1.59	1.87	1.89	1.89	1.89
ΔDEP	1.55	1.59	1.59	1.59	1.59

Note: Cholesky ordering: $\text{FFR} \rightarrow \text{OIL} \rightarrow \text{CPI} \rightarrow \text{DEP} \rightarrow \text{NEER} \rightarrow \text{VNI}$. Each row reports the percentage contribution of each structural shock to the forecast error variance of VN-Index returns at different horizons. All rows sum to 100%.

Index fluctuations is driven by factors not captured by the macroeconomic variables included in the VAR system.

Second, among the macroeconomic variables, FFR contributes the largest share to VN-Index variance, increasing modestly from 9.4% at $h = 1$ to around 10.0% at longer horizons. This finding indicates that external monetary policy shocks retain persistent explanatory power for Vietnamese equity market fluctuations.

Third, the FEVD converges rapidly and becomes largely stable after approximately six months, indicating that the dynamic effects of macroeconomic shocks are mostly absorbed within the short to medium run. This pattern is consistent with the impulse response analysis and the stability condition of the estimated VAR model.

Among the remaining variables, NEER and oil price shocks each explain approximately 4.5% of forecast error variance, while CPI and deposit rate shocks contribute less than 2% individually. Overall, the results suggest that external shocks play a relatively larger role than domestic macroeconomic variables in explaining VN-Index return variability over the forecast horizon.

V. DISCUSSION OF FINDINGS

The empirical results presented in Section 4 yield three principal findings. First, the Federal Funds Rate (FFR) is the only macroeconomic variable found to Granger-cause VN-Index returns.

Second, own shocks remain the dominant source of forecast error variance across all horizons, indicating the importance of market-specific dynamics and investor sentiment. Third, the domestic monetary policy channel, proxied by the deposit rate, exerts only a weak and statistically insignificant effect on equity returns.

A The Federal Funds Rate as the Dominant External Driver

The empirical findings indicate that the Federal Funds Rate (FFR) appears to be the most important observable external macroeconomic contributor to Vietnamese equity returns during the sample period. Specifically, ΔFFR is the only variable found to Granger-cause $\Delta \ln \text{VNI}_t$ at the 5% significance level ($F = 4.395$, $p = 0.014$), while FEVD results show that FFR shocks account for approximately 10% of VN-Index forecast error variance at medium- and long-run horizons.

It should be noted, however, that this 10% share represents the largest contribution among all observable macroeconomic variables. The dominant share of VN-Index variance, approximately 77%, remains attributable to idiosyncratic market-specific dynamics, as discussed in Section V.C. The impulse response analysis further reveals that an FFR shock generates an immediate negative adjustment in VN-Index returns after the initial impact period, although the effect fades rapidly within several months.

This finding is broadly consistent with the emerging market monetary spillover literature. While [3] document the equity risk premium channel for the U.S. market, the transmission mechanism to emerging markets operates primarily through international capital flows and global liquidity conditions. [8] argue that tighter U.S. monetary policy generates substantial spillover effects on emerging market financial systems because higher U.S. interest rates increase the relative attractiveness of USD-denominated assets. As a consequence, global investors rebalance their portfolios away from riskier emerging market assets toward safer U.S. financial instruments, generating capital outflows and downward pressure on emerging market equity prices.

The Vietnamese market is particularly vulnerable to this mechanism because of its increasing financial openness and dependence on foreign portfolio investment. As documented by [5], Vietnam's rapid integration into global financial markets has enhanced market liquidity and accelerated capital market development, but has also increased sensitivity to external financial shocks. Foreign investor behaviour in Vietnam is strongly linked to global risk appetite, USD liquidity conditions, and exchange rate expectations. During periods of global "risk-off" sentiment, foreign investors tend to reduce exposure to Vietnamese equities, amplifying domestic market volatility and weakening investor confidence in a market still dominated by retail participants.

The 2022–2023 Federal Reserve tightening cycle provides a clear illustration of this transmission mechanism, operating through two reinforcing channels. The first operated through foreign capital outflows: as the FFR rose aggressively from near-zero levels to above 5%, the narrowing interest rate differential between USD- and VND-denominated assets triggered sustained foreign net selling on the Vietnamese equity market, placing direct downward pressure on VN-Index valuations.

The second channel operated through the do-

mestic policy response to exchange rate pressure. To defend the USD/VND rate, the State Bank of Vietnam (SBV) intervened through large-scale foreign exchange sales and aggressive liquidity absorption via treasury bill issuance and open market operations, at peak intensity withdrawing hundreds of trillions of VND from the banking system in late 2022. The resulting liquidity squeeze caused overnight interbank rates to spike sharply, elevated funding costs across the financial system, and generated an additional layer of downward pressure on equity prices that compounded the direct effect of foreign outflows. The VN-Index ultimately fell by approximately one-third during 2022, reflecting not a single external shock but the compounding interaction between foreign portfolio rebalancing and domestically amplified financial tightening.

The short-lived nature of the impulse responses is also economically meaningful. Although VN-Index returns react rapidly to FFR shocks, the effect dissipates within a few months, suggesting that the effects of macroeconomic shocks are absorbed relatively quickly by the market. This pattern is consistent with the idea of rapid information incorporation and temporary portfolio rebalancing behaviour among international investors. Because financial markets typically anticipate Federal Reserve policy actions before official announcements, much of the adjustment is priced into asset valuations quickly. Moreover, domestic policy responses may partially offset external shocks. In Vietnam, the State Bank of Vietnam shifted toward monetary easing during 2023, helping stabilise liquidity conditions and mitigating the persistence of earlier tightening pressures.

Overall, the results suggest that Vietnamese equity returns are sensitive to changes in global monetary conditions, particularly U.S. monetary policy shocks, primarily through capital flow and exchange rate channels. However, the relatively modest magnitude and short-lived persistence of the impulse responses, combined with the dominant role of own shocks in the FEVD, imply that

external monetary shocks constitute an important but incomplete explanation of VN-Index dynamics, with market-specific sentiment and idiosyncratic factors playing the larger structural role.

B Weakness of the Domestic Monetary Policy Channel

The empirical results indicate that the domestic monetary policy channel exerts only a limited influence on Vietnamese equity returns. The deposit rate neither Granger-causes VN-Index returns nor contributes substantially to forecast error variance, accounting for only 1.6% of VN-Index variability at the 12-month horizon. Although the impulse response function suggests a negative response of equity returns to a deposit-rate shock, the effect remains statistically insignificant and dissipates rapidly. This finding reflects several structural characteristics of the Vietnamese financial system.

First, the weak transmission from deposit rates to equity valuation is consistent with the bank-based structure of Vietnam's financial system. Vietnam remains heavily dependent on bank credit, while capital markets continue to play a relatively secondary role in financial intermediation. In such systems, monetary policy tends to operate primarily through the lending and cost channels rather than through asset-price revaluation mechanisms. [9] show that monetary tightening in Vietnam is transmitted mainly through higher borrowing costs faced by firms, while the asset-price channel remains weak and statistically limited. Consequently, changes in deposit rates do not translate strongly into stock valuation adjustments in the same manner observed in more market-based financial systems such as the United States.

Second, the limited explanatory power of deposit rates may also reflect the behavioural characteristics of the Vietnamese stock market. Trading activity on HOSE is dominated by retail investors, whose portfolio decisions are often influenced more strongly by sentiment, speculative momen-

tum, and herd behaviour than by the traditional opportunity-cost mechanism implied by interest-rate theory. During the post-pandemic market surge of 2020–2021, for example, VN-Index increased substantially despite historically low deposit rates, suggesting that liquidity conditions and speculative optimism dominated conventional discount-rate effects. This interpretation is consistent with recent evidence that Vietnamese equity flows are highly sensitive to behavioural and global risk factors rather than purely domestic monetary variables.

Third, deposit rates may not fully capture the overall monetary stance of the State Bank of Vietnam (SBV). Unlike advanced economies that rely predominantly on market-based policy transmission, the SBV frequently employs administrative and macroprudential measures, including credit-growth quotas, liquidity guidance, and foreign-exchange interventions. During the 2022 tightening episode, for instance, policy tightening was implemented not only through higher interest rates but also through credit restrictions and tighter supervision of the corporate bond and real-estate sectors. Similarly, the market recovery in 2023 coincided not merely with policy rate cuts, but also with the relaxation of credit constraints and direct policy support measures [26]. As a result, the deposit rate alone provides only a partial proxy for Vietnam's broader monetary policy stance, helping explain its statistically insignificant effect in the VAR framework.

C The Dominance of Own Shocks: Sentiment and Market-Specific Dynamics

The FEVD results indicate that own shocks account for approximately 77.4% of VN-Index forecast error variance at the 12-month horizon, implying that the majority of stock market fluctuations in Vietnam originate from market-specific dynamics rather than from the observable macroeconomic variables included in the VAR system.

Although a relatively large idiosyncratic component is common in emerging and frontier equity markets [5], the magnitude observed in the present study suggests that investor sentiment, expectations, and endogenous market dynamics play a dominant role in driving VN-Index returns.

One explanation relates to the behavioural structure of the Vietnamese equity market. Trading activity on HOSE is dominated by retail investors, making the market more susceptible to herd behaviour, speculative momentum, and abrupt shifts in investor sentiment [26, 27]. In such market structures, price movements are often driven less by macroeconomic fundamentals and more by collective expectations regarding future market conditions. This behaviour became particularly visible during the COVID-19 period: despite severe economic disruption in 2020, the VN-Index recovered rapidly and reached record highs by early 2022, supported by an unprecedented surge in retail participation and speculative liquidity [26]. The episode illustrates how sentiment-driven trading can temporarily decouple equity prices from underlying macroeconomic conditions.

A second explanation is that standard macroeconomic variables may not fully capture the range of shocks affecting the Vietnamese equity market. Over the sample period, VN-Index dynamics were strongly influenced by market-specific and institutional events, including the COVID-19 pandemic, the 2022 corporate bond market stress, real-estate sector liquidity problems, and domestic banking concerns. In particular, the tightening of corporate bond regulations under Decree 65/2022 and the associated liquidity pressures in the real-estate sector generated substantial volatility in Vietnamese equities that cannot be adequately represented by aggregate macroeconomic indicators such as inflation, exchange rates, or deposit rates alone. As a result, a large portion of market variability remains embedded within the residual “own-shock” component of the VAR system.

Third, the dominance of own shocks is also

consistent with the characteristics of a frontier-style equity market with limited informational efficiency. Compared with more mature financial markets, Vietnamese equity prices are more sensitive to short-term capital flows, order imbalances, and speculative trading cycles. Lower market depth and thinner institutional participation reduce the speed and accuracy with which macroeconomic information is incorporated into asset prices. Consequently, market volatility tends to be amplified by endogenous feedback mechanisms, in which past price movements and investor psychology reinforce subsequent trading behaviour. The relatively rapid stabilisation of the FEVD after approximately six months further supports this view, suggesting that external macroeconomic shocks are absorbed fairly quickly, after which VN-Index dynamics revert primarily to internally generated market fluctuations [27].

D The Marginal Roles of NEER and Oil Price

NEER and oil price each explain approximately 4.5% of VN-Index forecast error variance at the 12-month horizon, making them secondary contributors relative to the dominant roles of FFR shocks and the market’s own innovations. Although both variables display weak and statistically insignificant impulse responses, their limited explanatory power remains economically interpretable within the institutional and structural context of Vietnam’s financial system.

The relatively small contribution of NEER shocks reflects the managed exchange rate regime maintained by the State Bank of Vietnam (SBV). Unlike freely floating emerging market currencies, the Vietnamese dong operates under a tightly managed crawling-peg framework in which the SBV publishes a daily central reference rate and constrains USD/VND trading within an administrative band, while actively deploying foreign exchange reserves to prevent movements from approaching band limits. As a result, monthly fluctuations in the

nominal effective exchange rate remain relatively small and predictable over most of the sample period, and the limited variability of $\Delta \ln(\text{NEER}_t)$ constrains the extent to which exchange rate shocks can propagate into equity valuations through the standard competitiveness or capital flow channels.

A key mechanism underlying this finding relates to the informational content of exchange rate movements under a managed regime. This institutional design fundamentally differs from the FFR transmission channel: whereas unexpected changes in U.S. monetary policy generate genuine surprise shocks to global risk appetite and capital flows, NEER movements in Vietnam are largely anticipated by market participants, precisely because the SBV's intervention smooths and signals exchange rate adjustments in advance. As a result, monthly fluctuations in $\Delta \ln(\text{NEER}_t)$ carry limited new information for equity pricing – the market has already incorporated expected exchange rate movements into asset valuations before they are realised in the data. Under the efficient market hypothesis [11], only the unexpected component of any macroeconomic variable should generate equity price responses. By systematically reducing the surprise component of exchange rate changes, the SBV's managed regime effectively reduces the pricing relevance of NEER shocks for Vietnamese equity returns. This anticipation mechanism, rather than the absence of exchange rate effects per se, helps explain why NEER shocks contribute only modestly to VN-Index forecast error variance despite exchange rate pressures being economically significant during episodes such as the 2022–2023 tightening cycle.

This finding is consistent with [9, 28], who similarly document a weak exchange rate transmission channel in Vietnam. In a heavily managed monetary system, exchange rate movements do not fully reflect market-clearing adjustments to external shocks, thereby reducing their informational content for equity pricing. Consequently, exchange rate dynamics play only a modest role in explaining

VN-Index fluctuations relative to financial sentiment and global monetary conditions.

The marginal contribution of oil price shocks can also be explained by the heterogeneous sectoral structure of the Vietnamese equity market. Rising oil prices benefit energy-related firms through higher revenues and profit margins, while simultaneously increasing production and transportation costs for manufacturing, logistics, and consumer-oriented sectors. These opposing sectoral effects partially offset one another at the aggregate index level, weakening the net impact of oil price changes on VN-Index returns.

This interpretation is consistent with the broader evidence for emerging equity markets reported by [20], who finds that oil price shocks have limited explanatory power for stock market returns in developing economies. In the Vietnamese context, oil price movements appear to influence equity prices primarily through indirect channels – such as inflation, exchange rate pressures, and production costs – rather than through a strong direct transmission mechanism to aggregate stock valuations.

More broadly, the weak roles of both NEER and oil price reinforce the central finding of this study: VN-Index dynamics are driven predominantly by financial and behavioural forces rather than by conventional domestic macroeconomic fundamentals. Compared with the dominant effects of FFR shocks, investor sentiment, and the market's own internal dynamics, external macroeconomic variables such as exchange rates and oil prices play only marginal roles in explaining fluctuations in the Vietnamese stock market over the 2010–2024 period.

E Robustness and Model Consistency (ARDL vs VAR)

The consistency between the VAR and ARDL specifications provides important evidence that the empirical findings are not driven primarily by

model-specific assumptions or estimation procedures. While the VAR framework captures dynamic interactions within a system-based setting, the ARDL specification offers a complementary single-equation representation of short-run adjustment dynamics. The broadly consistent coefficient signs and direction of statistical significance across both models – particularly the persistent negative effect of FFR – suggest that the identified transmission mechanisms are qualitatively robust to specification choice, even if point estimates differ across frameworks due to differences in data transformation and model structure.

From an econometric perspective, the robustness of the results indicates that the core qualitative relationships between VN-Index returns and macroeconomic variables are not highly sensitive to specification choice. In particular, the Federal Funds Rate maintains a consistently negative coefficient across both models, suggesting that tighter global monetary conditions exert persistent downward pressure on Vietnamese equity returns even when individual coefficient significance varies across specifications. By contrast, domestic monetary variables – particularly the deposit rate – remain economically weak and statistically unstable in both frameworks, reinforcing the conclusion that the domestic interest rate transmission channel to equity valuations is limited.

The similarity between the VAR and ARDL estimates also supports the interpretation that short-run market dynamics dominate VN-Index fluctuations over the sample period. Although macroeconomic fundamentals retain some explanatory power, the relatively weak and unstable coefficients associated with domestic variables suggest that market-specific dynamics, investor sentiment, and external financial conditions play substantially larger roles in determining short-run equity returns.

Overall, the consistency across alternative specifications strengthens the credibility of the empirical findings and reduces concerns regarding model dependence in the macro-financial analysis of the

Vietnamese stock market.

VI. CONCLUSION

This study investigates the macroeconomic determinants of VN-Index returns over the period January 2010 to December 2024 using a six-variable VAR(2) framework with structural identification based on Cholesky decomposition. The sample period encompasses significant shifts in both domestic and global financial conditions, most notably the 2022–2023 Federal Reserve tightening cycle, which represents the most aggressive U.S. monetary policy adjustment in four decades and constitutes a key episode of external shock transmission examined in this study.

Three principal findings emerge from the empirical results. First, among the macroeconomic variables considered, only the U.S. Federal Funds Rate (FFR) is found to Granger-cause VN-Index returns at the 5% significance level. Orthogonalized impulse response analysis further shows that an unexpected tightening in U.S. monetary policy generates a statistically significant negative response in VN-Index returns in the short run, although the effect dissipates relatively quickly. This finding suggests that global monetary conditions constitute the most important identifiable external transmission channel affecting the Vietnamese equity market. In particular, changes in the FFR appear to influence Vietnamese equity valuations primarily through global capital flows, risk rebalancing behaviour, and shifts in investor sentiment.

Second, forecast error variance decomposition indicates that VN-Index dynamics are dominated by idiosyncratic or market-specific shocks. At the 12-month horizon, own shocks account for approximately 77.4% of forecast error variance, while FFR shocks contribute roughly 10%, representing the largest share among all macroeconomic variables. Domestic macroeconomic indicators, including CPI, deposit rate, NEER, and oil price, explain only

a relatively limited portion of VN-Index variability. These results imply that the Vietnamese stock market remains strongly influenced by sentiment-driven behaviour, speculative trading activity, and institutional or market-specific events, consistent with the characteristics commonly observed in emerging and frontier equity markets.

Third, the empirical evidence suggests that the domestic monetary policy transmission channel to equity prices remains relatively weak. Although deposit-rate shocks produce statistically insignificant and economically modest responses, the mixed sign pattern across horizons is consistent with the limited effectiveness of interest-rate transmission in Vietnam's bank-dominated financial structure. This outcome reflects the continued importance of bank credit allocation, administrative policy measures, and the comparatively underdeveloped role of asset-price channels in transmitting monetary policy to equity valuations. Robustness checks using an ARDL specification confirm sign consistency across all key coefficients, supporting the stability of the main empirical conclusions.

Taken together, these findings position the Vietnamese equity market as one whose short-run dynamics are shaped more by external financial conditions and market-specific sentiment than by domestic macroeconomic fundamentals – a pattern consistent with the broader literature on frontier and emerging market equity behaviour.

Several limitations of the present study provide directions for future research. First, the linear VAR framework assumes constant coefficients throughout the sample period and may therefore overlook structural changes associated with major episodes of global financial stress, such as the 2022–2023 Fed tightening cycle. Future studies could employ time-varying parameter VAR models or regime-switching frameworks to examine whether macro-financial transmission mechanisms evolve over time. Second, the large share of unexplained VN-Index variance suggests that important determinants such as foreign capital flows, domestic

credit conditions, and investor sentiment remain outside the current specification. Incorporating such variables into an extended framework may improve the explanatory power of future research. Finally, future studies may also explore non-linear methodologies and higher-frequency financial data to better capture the behavioural and speculative dynamics characteristic of the Vietnamese equity market.

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